
Tools, Not Talk: Agent Orchestration and Extended Reasoning Hit a Knowledge Ceiling that Tool Use Breaks on Biology Research Tasks

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Abstract

1 Agentic systems for biology usually report gains over a single model call while
2 spending several calls, which conflates what the orchestration adds with how much
3 inference is spent. On 150 pre-registered, deterministically scored biology re-
4 search questions (LAB-Bench, LABBench2), we compare majority voting, self-
5 critique and a two-scientist debate at a matched budget of three calls across three
6 models, and we contrast them with longer reasoning and with tool use. Critique
7 never beats voting, and debate helps only a model whose reasoning is switched
8 off (+8.7 points). Raising reasoning effort rarely helps (GPT-OSS spends $25\times$
9 more tokens for no gain). Instead, correctness is set by the question: question
10 identity explains 56% of its variance and the choice of model, strategy and effort
11 0.6%. The ceiling is missing knowledge, not missing deliberation: a single call
12 with web search and code execution solves 47% of the questions that every or-
13 chestrated or reasoning-heavy system failed (vs. 6% for the same model without
14 tools), and more compute on top of tools adds only a further, non-significant +6
15 to +13 points. For life-science agents, the budget belongs in tools, not in talk.

16 1 Introduction

17 LLM agents now plan experiments, analyse data and generate hypotheses in chemistry and biology
18 [1, 2, 5, 6, 8, 16]. Their gains are typically reported against a single-call baseline, although the agent
19 makes many calls. Because repeated sampling with a vote already improves accuracy [3, 12, 19],
20 such comparisons conflate *what the interaction adds* with *how much inference was spent* [9].

21 For anyone building life-science agents, this is a design question as much as an evaluation one:
22 should the budget go into richer orchestration (critics, debating scientists) or somewhere else? We
23 run a small, controlled test of this confound on biology research questions with deterministic an-
24 swers. Every orchestrated strategy gets exactly the budget of the voting baseline (three calls, same
25 model, same prompts, same decoding). We then ask a second question the agent literature rarely
26 asks: does letting a strong model reason longer in a *single* call reach the same point?

27 **Findings.** (1) At matched budget, sequential critique never improves on majority voting, and
28 multi-agent debate improves on it only for a model whose reasoning is switched off. (2) Critique
29 never helps: it breaks at least as many correct answers as it repairs, and for models that reason it is
30 most harmful on exact sequence manipulation. (3) Thinking longer is not a way out: higher reason-
31 ing effort helps only a model that otherwise does not reason at all, and then only up to the level its
32 low-effort debate already reached; stacking orchestration on top adds nothing reliable. (4) Whether
33 a system is right depends overwhelmingly on *which question* it faces and barely on *how* it spends
34 inference: every system solves and fails the same questions (Fig. 2). (5) These errors are missing

Table 1: Accuracy (%) at a matched budget of three calls; DIRECT uses one call. Δ is DEBATE – VOTE in points with a paired bootstrap 95% CI.

Model	n	DIRECT	VOTE	CRITIQUE	DEBATE	Δ [95% CI]
GPT-OSS-120B	150	46.7	49.3	42.7	47.3	-2.0 [-8.7, +4.7]
Gemma 4 31B	150	48.7	48.7	47.3	57.3	+8.7 [+2.0, +15.3]
Flash-Lite	75	61.3	62.7	56.0	57.3	-5.3 [-13.3, +1.3]

35 knowledge, not missing deliberation. One tool-using call with web search and code execution solves
 36 nearly half of the questions that every other configuration failed, and extra compute on top of tools
 37 adds comparatively little.

38 2 Setup

39 **Tasks.** We use four LAB-Bench categories [10] (ProtocolQA, SeqQA, DbQA, LitQA2; multiple
 40 choice with shuffled options and no abstain option) and the deterministic subset of LABBench2
 41 SeqQA2 [11] (open-ended sequence computations with unique answers, scored by the official val-
 42 idators, with input files injected into the prompt). We exclude figure/table tasks (text-only models)
 43 and prompts over 6,000 characters. With a fixed seed we draw 30 questions per category (150 total),
 44 plus 10 pilot questions that are excluded from all results. No LLM judge is used anywhere.

45 **Strategies (Fig. 3 in App. A).** DIRECT makes one call. VOTE makes three independent calls and
 46 takes the majority answer, breaking ties by self-reported confidence. CRITIQUE solves, critiques the
 47 solution, then revises. DEBATE has a molecular biologist and a skeptical reviewer solve indepen-
 48 dently, and a third call adjudicates between them. VOTE, CRITIQUE and DEBATE each use exactly
 49 three calls. Only visible outputs are passed between calls.

50 **Models.** The main study uses GPT-OSS-120B [13] (`reasoning_effort=low`), Gemma 4 31B
 51 [17] (thinking off, the only non-default setting the API accepts), and Gemini 3.5 Flash-Lite (thinking
 52 low), all through free API tiers. Decoding uses temperature 0.6, fixed per-call seeds and a 2,048-
 53 token cap. GPT-OSS and Gemma answer all 150 questions and Flash-Lite a fixed 75-question
 54 subset. For the reasoning-effort analysis we additionally run GPT-6 Luna, GPT-6 Sol and GPT-5.6
 55 Terra (all strategies, high effort; Luna also at low effort), GPT-OSS at high effort (all strategies,
 56 served by Ollama; a low-effort DIRECT run on the same host matches the main run’s accuracy)
 57 and Flash-Lite (DIRECT at high thinking), all on the first 30 questions in the fixed run order (6 per
 58 category).

59 **Pre-registration.** Task IDs, prompts, decoding, hypotheses and the analysis script were commit-
 60 ted before the main run. The primary contrast is DEBATE – VOTE, tested with a paired bootstrap
 61 over questions (10,000 resamples) and an exact McNemar test. Post-freeze deviations are listed
 62 in App. B: rate-limit workarounds, a second inference provider for 22 GPT-OSS questions (each
 63 question served entirely by one provider), and the exploratory reasoning-effort arm.

64 3 Results

65 **Orchestration beats voting only when the model cannot think.** DEBATE improves on VOTE for
 66 Gemma, whose thinking is disabled (+8.7 points, 95% CI [+2.0, +15.3], McNemar $p = 0.015$),
 67 and the gain survives excluding every question with a truncated call. For the two models that reason,
 68 the interval includes zero ($p = 0.7$ and 0.29 ; for GPT-OSS the estimate is unchanged at -1.6 when
 69 the 22 questions served by the second provider are excluded). Pooled over all 375 model–question
 70 pairs, DEBATE – VOTE is $+1.6$ [$-2.4, +5.6$] points and CRITIQUE – VOTE is -4.5 [$-9.3, +0.3$].
 71 CRITIQUE is never better than VOTE ($-6.7, -1.3, -6.7$ points). Voting itself barely improves on
 72 a single call ($+2.7, +0.0, +1.3$ points), even though the three samples disagree on 38%, 42% and
 73 27% of questions.

74 **Critique breaks at least as often as it repairs.** Relative to the vote, CRITIQUE rescues 24%, 13%
 75 and 25% of wrong answers but corrupts 38%, 16% and 26% of correct ones. For the two models

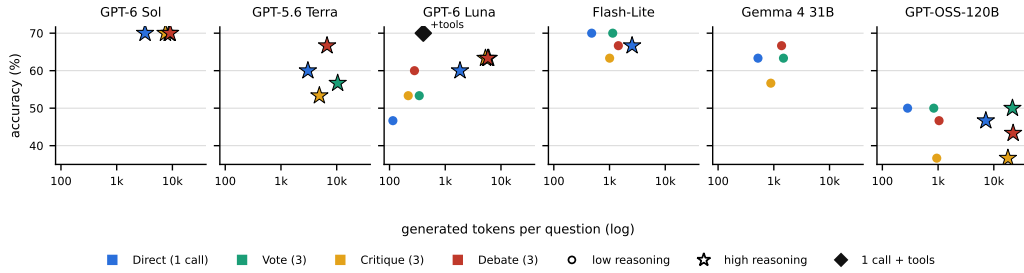


Figure 1: Accuracy vs. generated tokens per question on the same 30 questions. Circles are low reasoning and stars are high reasoning; colour marks the strategy. Across nearly two orders of magnitude in generated tokens, all ways of spending inference land on a narrow per-model plateau; the black diamond (GPT-6 Luna, one call with web search and code execution) is the only configuration that rises above it, at a fraction of the tokens.

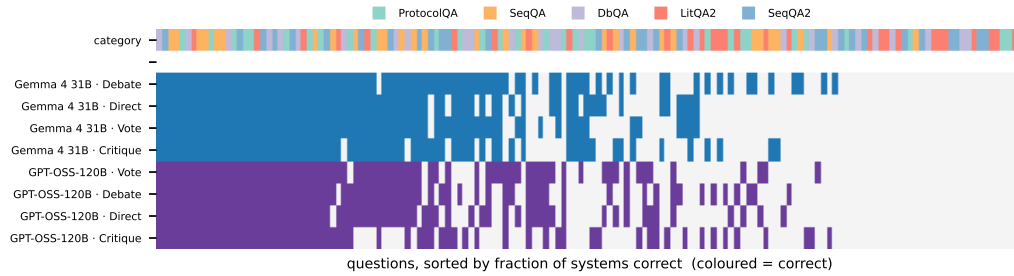


Figure 2: Correctness of every model $\times$ strategy system (rows) on every question answered by all of them (columns, sorted by the fraction of systems correct). The staircase shows that systems solve and fail the *same* questions and differ mainly in a shared boundary. Question identity explains 56% [49, 62] of the variance in correctness; the system explains 0.6% [0.3, 1.9].

76 that reason, the largest losses are on exact sequence manipulation (SeqQA: -23 and -20 points vs.
77 VOTE for GPT-OSS and Flash-Lite; +10 for Gemma), where a fluent but wrong critique talks the
78 model out of a correct computation.

79 **Thinking longer mostly does not help either.** On the same 30 questions (Fig. 1), raising GPT-
80 OSS to high reasoning effort multiplies generated tokens by $25\times$ (284 to 7,191 per question) while
81 accuracy stays flat (50% to 47%). For Flash-Lite the result is the same (70% to 67%, $5\times$ tokens).
82 The one model that gains is GPT-6 Luna, which at low effort uses no reasoning tokens (47% to
83 60%). Its high-effort single call lands exactly where its low-effort DEBATE already was (60%),
84 consistent with debate acting as a substitute for reasoning. Orchestration on top of high effort adds
85 nothing reliable. GPT-6 Sol scores 70% with all four strategies, and GPT-5.6 Terra ranges from 53%
86 (CRITIQUE) to 67% (DEBATE) with no paired difference significant. Within each model, accuracy
87 moves by a few questions while generated tokens vary by up to two orders of magnitude.

88 **The question, not the method, determines the outcome.** A two-way decomposition of the 0/1
89 correctness matrix (systems $\times$ questions) attributes 56% of the variance to the question and 0.6%
90 to the system for the main models (Fig. 2). Across all 33 systems on the shared 30 questions,
91 including frontier models at high reasoning, the split is 47% vs. 3.9%. Of the 32 questions that no
92 main system solves, 18 require literature or database recall (LitQA2, DbQA) and 9 are open-ended
93 sequence computations (SeqQA2), failures that point to missing retrieval and code execution rather
94 than missing deliberation.

95 **Tools break the ceiling.** To test whether the ceiling reflects missing knowledge, we give GPT-6
96 Luna (low effort, one call) a shell with Python and live web search, announced by a one-sentence
97 harness note, and leave everything else unchanged. On the shared 30 questions accuracy rises from
98 47% to 70% (+23 points, 95% CI [+7, +43], McNemar $p = 0.039$), more than any orchestration or
99 reasoning setting, while generating only ~ 404 tokens per question. On the 32 questions that none of

the eight main systems solved, it reaches 47% vs. 6% for the same model without tools (+41 points [+25, +56], $p < 0.001$). The gain comes from retrieval: LitQA2 rises from 18% to 91% and DbQA from 0% to 43% on these questions, whereas code execution barely moves the open-ended SeqQA2 computations (0% to 11%). That retrieval helps on literature and database questions is expected (these categories were designed for retrieval agents [10, 14]); what matters here is the contrast with every orchestration and reasoning setting, none of which moved these questions. Compute on top of tools helps modestly: high effort reaches 83% on the shared questions (+13 points over low-effort tools, $p = 0.22$) and a three-call vote 77% (+7, $p = 0.73$), at $\sim 6\times$ and $\sim 3.5\times$ the tokens. All tool calls were logged; no search or command reached the benchmark or its datasets.

4 Implications for building life-science agents

Benchmark against a compute-matched vote. A multi-call agent should beat independent sampling with the same number of calls, not a single call [9]. Most orchestration gains vanish under that baseline. **Spend the harness budget on tools, not talk.** Critics, debaters and longer reasoning re-deliberate over the same missing knowledge. Retrieval over literature and databases supplies it, at a fraction of the tokens. **Sequence computation needs better tools than a bare shell.** Generic code execution rarely fixed the open-ended sequence tasks, which suggests that domain tool ecosystems (sequence libraries, primer and cloning tools) and specialist models are where the next gains lie. Our results agree with compute-matched analyses of debate on general reasoning benchmarks [4, 15, 18, 20] and with evidence that intrinsic self-correction is unreliable [7]. We extend them to biology, to sequential critique, and to the trade-off between orchestration and reasoning effort.

Limitations. Models had no tools or retrieval, so DbQA and LitQA2 measure recall; orchestration with tools may behave differently. The budget is fixed at three calls. Frontier models were run through the Codex CLI, which adds its own system prompt and does not expose temperature or seeds; they cover 30 questions, so their comparisons are exploratory. Gemma could only run with thinking off. One prompt set per strategy was used, and 150 questions give limited power for effects under ~ 10 points. The tools arm uses one model, a harness note that the no-tools condition lacks, and 30+32 questions; web retrieval can in principle surface benchmark-derived pages, which our logs did not show.

5 Conclusion

On objectively scored biology research tasks, how a model spends extra inference (voting, self-critique, multi-agent debate or longer reasoning) barely changes which questions it gets right: question identity explains 56% of the variance in correctness and the choice of system 0.6%. The ceiling is knowledge, not deliberation, and a single tool-using call breaks it where no amount of orchestration did. Builders of life-science agents should spend their budget on tools, not talk.

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201 A Prompts

202 All prompts are single user messages. {QUESTION} is the question (with protocol, options or in-
203 jected files) and {ANSWER INSTRUCTION} is the format line for multiple choice or the official Se-
204 qQA2 suffix.

205 Direct / Independent×3 / Critique call 1 (solve)

206 Solve the following biology research question.
207
208 {QUESTION}
209
210 {ANSWER INSTRUCTION}
211
212 Respond in exactly this format and keep it short:
213 <answer>YOUR ANSWER</answer>
214 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
215 JUSTIFICATION: <at most two sentences>

216 Critique call 2 (critique)

217 Below are a biology research question and a proposed solution. Act as a critical reviewer: identify any
218 scientific, computational or reasoning error in the solution. If you think the answer is wrong, say which
219 answer you believe is correct and why. If you find no error, say so.
220
221 {QUESTION}
222
223 Proposed solution:
224 {CALL 1 OUTPUT}
225
226 Respond with at most four sentences, starting with "CRITIQUE:".

227 Critique call 3 (revise)

228 Below are a biology research question, a first attempt at solving it, and a reviewer's critique of that
229 attempt. Either may contain mistakes. Considering both, produce the final answer.
230
231 {QUESTION}
232
233 First attempt:
234 {CALL 1 OUTPUT}
235
236 Reviewer's critique:
237 {CALL 2 OUTPUT}
238
239 {ANSWER INSTRUCTION}
240
241 Respond in exactly this format and keep it short:
242 <answer>YOUR ANSWER</answer>
243 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
244 JUSTIFICATION: <at most two sentences>

245 Debate call 1 (researcher)

246 You are a molecular biology researcher. Solve the following biology research question independently.
247
248 {QUESTION}
249
250 {ANSWER INSTRUCTION}
251
252 Respond in exactly this format and keep it short:
253 <answer>YOUR ANSWER</answer>
254 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
255 JUSTIFICATION: <at most two sentences>

256 Debate call 2 (skeptic)

257 You are a skeptical scientific reviewer. Solve the following biology research question independently, looking
 258 especially for subtle experimental, biological or computational errors and traps.
 259
 260 {QUESTION}
 261
 262 {ANSWER INSTRUCTION}
 263
 264 Respond in exactly this format and keep it short:
 265 <answer>YOUR ANSWER</answer>
 266 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
 267 JUSTIFICATION: <at most two sentences>

268 Debate call 3 (adjudicator)

269 Two scientists independently analysed the biology research question below. Adjudicate between their analyses
 270 (either or both may be wrong) and submit the final answer.

271
 272 {QUESTION}
 273
 274 Analysis 1 (molecular biology researcher):
 275 {CALL 1 OUTPUT}
 276
 277 Analysis 2 (skeptical scientific reviewer):
 278 {CALL 2 OUTPUT}
 279
 280 {ANSWER INSTRUCTION}
 281
 282 Respond in exactly this format and keep it short:
 283 <answer>YOUR ANSWER</answer>
 284 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
 285 JUSTIFICATION: <at most two sentences>

286 B Pre-registration deviations

287 All task IDs, prompts, decoding settings and the primary analysis were frozen after a 10-question
 288 pilot and before any main-set call. Changes made after the freeze: (i) throttle and retry fixes for an
 289 undocumented 150 requests/hour provider limit (no completed call was re-run); (ii) 22 GPT-OSS
 290 questions served in full by a second provider of the same model (identical prompts, seeds and cap;
 291 excluding them leaves DEBATE – VOTE at -1.6), with partially served questions discarded and
 292 re-run on the primary provider; (iii) the exploratory reasoning-effort arm on the first 30 questions,
 293 including frontier models served through the Codex CLI with web search, code execution and sub-
 294 agents disabled (a first attempt with web search enabled was discarded unanalysed), and GPT-OSS
 295 at high effort on a second host.

296 C Additional results

Category	Gemma					Lite					OSS				
	D	I	C	M	Disagree	D	I	C	M	Disagree	D	I	C	M	Disagree
DbQA	56.7	53.3	40.0	56.7	33.3	40.0	40.0	40.0	33.3	26.7	46.7	46.7	43.3	36.7	36.7
LitQA2	33.3	30.0	33.3	36.7	46.7	60.0	53.3	53.3	46.7	26.7	20.0	30.0	40.0	36.7	43.3
ProtocolQA	56.7	56.7	60.0	70.0	20.0	66.7	80.0	86.7	73.3	26.7	56.7	50.0	43.3	46.7	33.3
SeqQA	56.7	56.7	66.7	73.3	46.7	73.3	73.3	53.3	66.7	13.3	63.3	66.7	43.3	60.0	33.3
SeqQA2	40.0	46.7	36.7	50.0	63.3	66.7	66.7	46.7	66.7	40.0	46.7	53.3	43.3	56.7	43.3

Table 2: Per-category accuracy (D=Direct, I=Independent×3, C=Critique, M=Multi). Numbers are percentages.